Math 55 quiz 2/10/2026

You must type or neatly handwrite each answer on a separate piece of paper and select the pages for each question when you upload to Gradescope. Feel free to use more than one sheet for an answer if needed. Note that both PDF and picture/photo format (JPG) are accepted. You must show your work. During the quiz, you may consult the textbook, your own notes and other course materials you yourself prepared before the beginning of the quiz. You may not consult resources online. You can post a private question to Ed Discussion to instructors only, but do not make public posts on Ed Discussion, Discord, or other online forums.

By submitting this quiz, you agree to adhere to the statement below: "As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others. All of the work submitted here is mine, and I have not discussed the content of this exam with any other person. I have not consulted any resources beyond the course materials and my notes during the exam. I have not used calculators, computers, or the internet to do calculations or look up answers."

1. Prove that the sets $(0, 1)$ and $\mathbb{R}$ have the same cardinality. (Hint: Consider trig functions.)

Solution: It's enough to define a function $f(x) : (0, 1) \to \mathbb{R}$ which is a bjijection. We can use $f(x) = \tan(\pi x - \frac{\pi}{2})$. This function is well-defined because tangent takes finite values on every number in the interval $(-\pi/2, \pi/2)$. The function $f(x)$ is a bijection because it has a two-sided inverse given by $g(y) = \frac{2}{\pi}(\arctan(y) + \frac{\pi}{2})$. (Recall that being a two-sided inverse means that $g(f(x)) = x$ and $f(g(y)) = y$.)

2. Show that the last digit of every number greater than 5 and not divisible by 2 or 5 is either 1, 3, 7, or 9.

Solution: Let p be greater than 5. By the division algorithm, we can find q, r s.t. $p = 10q + r$. Note that r is the last digit of p.

Assume towards contradiction that r is not either $1, 3, 7$, or 9. Then, either r is $0, 2, 4, 6, 8$, or 5. For the cases of $0, 2, 4, 6$, and 8, note that we can factor out a multiple of 2; hence r is divisible by 2. For the case of 5, we can factor out a multiple of 5; thus r is divisible by 2.